

Answer ALL questions. ALL working should be shown.

1. Frame $\{B\}$ is initially coincident with frame $\{A\}$. First, $\{B\}$ is rotated about x_B by 30 degrees, and is subsequently rotated about y_B by 45 degrees, followed by a rotation about z_B by 15 degrees.

- (a) Show that the rotation matrix which accomplishes these 3 rotations in the given order is

$${}^A_B R = \begin{bmatrix} 0.6830 & -0.1830 & 0.7071 \\ 0.5657 & 0.7450 & -0.3536 \\ -0.4620 & 0.6414 & 0.6123 \end{bmatrix}$$

[5 marks]

The rotations occur about the axis of the rotated frame, thus this is Euler angles representation. For Euler angles, the sequence of individual rotation matrices is the same as the sequence of physical rotation (X then Y then Z). Therefore, the rotation matrix is as follows:

$$\begin{aligned} {}^A_B R &= R_x(30) \cdot R_y(45) \cdot R_z(15) \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & c30 & -s30 \\ 0 & s30 & c30 \end{bmatrix} \cdot \begin{bmatrix} c45 & 0 & s45 \\ 0 & 1 & 0 \\ -s45 & 0 & c45 \end{bmatrix} \cdot \begin{bmatrix} c15 & -s15 & 0 \\ s15 & c15 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0.866 & -0.5 \\ 0 & 0.5 & 0.866 \end{bmatrix} \cdot \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0 & 1 & 0 \\ -0.7071 & 0 & 0.7071 \end{bmatrix} \cdot \begin{bmatrix} 0.9659 & -0.2588 & 0 \\ 0.2588 & 0.9659 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.7071 & 0 & 0.7071 \\ 0.3536 & 0.866 & -0.3536 \\ -0.6123 & 0.5 & 0.6123 \end{bmatrix} \cdot \begin{bmatrix} 0.9659 & -0.2588 & 0 \\ 0.2588 & 0.9659 & 0 \\ 0 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0.6830 & -0.1830 & 0.7071 \\ 0.5657 & 0.7450 & -0.3536 \\ -0.4620 & 0.6414 & 0.6123 \end{bmatrix} \end{aligned}$$

[2 marks for the first 2 lines of equation, 3 marks for the numerical calculation.]

- (b) Calculate the Euler Parameters for the above rotation matrix.

[3 marks]

The Euler Parameters are:

$$\varepsilon_4 = \frac{1}{2} \sqrt{1 + r_{11} + r_{22} + r_{33}} = \frac{1}{2} \sqrt{1 + 0.6830 + 0.7450 + 0.6123} = \frac{1}{2} \sqrt{3.0403} = 0.8718.$$

$$\varepsilon_1 = \frac{r_{32} - r_{23}}{4\varepsilon_4} = \frac{0.6414 - (-0.3536)}{4(0.8718)} = 0.2853$$

$$\varepsilon_2 = \frac{r_{13} - r_{31}}{4\varepsilon_4} = \frac{0.7071 - (-0.4620)}{4(0.8718)} = 0.3353$$

$$\varepsilon_3 = \frac{r_{21} - r_{12}}{4\varepsilon_4} = \frac{0.5657 - (-0.1830)}{4(0.8718)} = 0.2147$$

[1.5 marks for ε_4 , 0.5 marks each for $\varepsilon_1, \varepsilon_2, \varepsilon_3$.]

- (c) Calculate the unit vector and the single axis of rotation (in equivalent angle-axis representation) for the above rotation matrix.

[2 marks]

The unit vector and single axis of rotation are:

$$\theta = \cos^{-1} \left(\frac{r_{11} + r_{22} + r_{33} - 1}{2} \right) = \cos^{-1} \left(\frac{0.6830 + 0.7450 + 0.6123 - 1}{2} \right) = 58.6577^\circ$$

$$\hat{K} = \frac{1}{2 \sin \theta} \begin{bmatrix} r_{32} - r_{23} \\ r_{13} - r_{31} \\ r_{21} - r_{12} \end{bmatrix} = \frac{1}{1.7081} \begin{bmatrix} 0.6414 - (-0.3536) \\ 0.7071 - (-0.4620) \\ 0.5657 - (-0.1830) \end{bmatrix} = \begin{bmatrix} 0.5825 \\ 0.6844 \\ 0.4383 \end{bmatrix}$$

[1 mark for θ , 1 mark for $\hat{K}$]

- (d) The origin of frame $\{B\}$ is then shifted such that it is at ${}^A P_{BORG} = [5 \ 10 \ 2]^T$. Calculate ${}^B_A T$.
[5 marks]

The homogeneous transformation matrix from $\{A\}$ to $\{B\}$ is:

$${}^A_B T = \begin{bmatrix} {}^A_B R & {}^A P_{BORG} \\ 0 \ 0 \ 0 & 1 \end{bmatrix} = \begin{bmatrix} 0.6830 & -0.1830 & 0.7071 & 5 \\ 0.5657 & 0.7450 & -0.3536 & 10 \\ -0.4620 & 0.6414 & 0.6123 & 2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

${}^B_A T$ is the inverse of ${}^A_B T$, and can be calculated as follows:

$${}^B_A T = {}^A_B T^{-1} = \begin{bmatrix} {}^A_B R^T & -{}^A_B R^T \cdot {}^A P_{BORG} \\ 0 \ 0 \ 0 & 1 \end{bmatrix}$$

where:

$${}^A_B R^T = \begin{bmatrix} 0.6830 & 0.5657 & -0.4620 \\ -0.1830 & 0.7450 & 0.6414 \\ 0.7071 & -0.3536 & 0.6123 \end{bmatrix}$$

$$-{}^A_B R^T \cdot {}^A P_{BORG} = \begin{bmatrix} -0.6830 & -0.5657 & 0.4620 \\ 0.1830 & -0.7450 & -0.6414 \\ -0.7071 & 0.3536 & -0.6123 \end{bmatrix} \cdot \begin{bmatrix} 5 \\ 10 \\ 2 \end{bmatrix} = \begin{bmatrix} -8.148 \\ -7.8178 \\ -1.2241 \end{bmatrix}$$

Therefore,

$${}^B_A T = {}^A_B T^{-1} = \begin{bmatrix} 0.6830 & 0.5657 & -0.4620 & -8.148 \\ -0.1830 & 0.7450 & 0.6414 & -7.8178 \\ 0.7071 & -0.3536 & 0.6123 & -1.2241 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

[1 mark for ${}^A_B T$, 1 mark for ${}^A_B R^T$, 2 marks for $-{}^A_B R^T \cdot {}^A P_{BORG}$, 1 mark for putting together into ${}^B_A T$.]

2. The kinematic structure of a 3-link RPP robot is shown in Figure 2.1. The x_0 axis is given, and the configuration as shown in Figure 2.1 is when $\theta_1 = 0^\circ$.

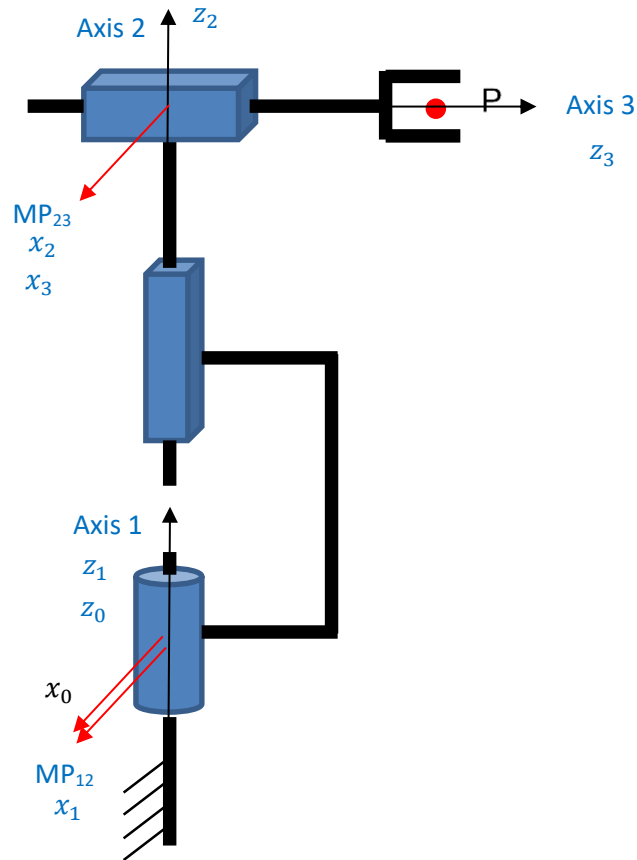


Figure 2.1: RPP robot

- (a) Derive the DH-parameters of the robot, and show that the transformation matrix from frame $\{0\}$ to frame $\{3\}$ is:

$${}^0_3T = \begin{bmatrix} c_1 & 0 & -s_1 & -d_3 \cdot s_1 \\ s_1 & 0 & c_1 & d_3 \cdot c_1 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Note:

- If there is any ambiguity in up-down direction, choose up.
- If there is any ambiguity in left-right direction, choose right.
- If there is any ambiguity in into the plane / out of the plane, choose out of the plane.

[10 marks]

Step 1: Draw axes (directly on Figure 2.1).

Step 2: Draw mutual perpendicular (MP) from lower axis to higher axis. Note: MP from axis 1 to axis 2 could be arbitrary. However, since x_0 is given, and since $\theta_1 = 0^\circ$ in the configuration as shown, MP_{12} must be the same as x_0 .

Step 3: Draw frames $\{1\}$ and $\{2\}$. Z axis aligned with rotary/translation, where X axis aligned with mutual perpendiculars.

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Step 4: Draw frame {0}. In this case, x_0 is already given. We only need to draw z_0 aligned with z_1 .

Step 5: Draw frame {3}. Z axis aligned with axis 3, X axis aligns with x_2 .

Step 6: Determine link lengths a_{i-1} :

a_0 = distance from z_0 to z_1 along $x_0 = 0$.

a_1 = distance from z_1 to z_2 along $x_1 = 0$.

a_2 = distance from z_2 to z_3 along $x_2 = 0$.

Step 7: Determine link twists α_{i-1} :

α_0 = angle between z_0 and z_1 about $x_0 = 0$.

α_1 = angle between z_1 and z_2 about $x_1 = 0$.

α_2 = angle between z_2 and z_3 about $x_2 = -90^\circ$.

Step 8: Determine link offsets d_i :

d_1 = distance from x_0 to x_1 along $z_1 = 0$.

d_2 = distance from x_1 to x_2 along $z_2 = \text{variable}$.

d_3 = distance from x_2 to x_3 along $z_3 = \text{variable}$.

Step 9: Determine joint angles θ_i :

θ_1 = angle between x_0 and x_1 about $z_1 = \text{variable}$.

θ_2 = angle between x_1 and x_2 about $z_2 = 0$.

θ_3 = angle between x_2 and x_3 about $z_3 = 0$.

Step 10: Summarize into a DH Table:

i	α_{i-1}	a_{i-1}	d_i	θ_i
1	0	0	0	θ_1
2	0	0	d_2	0
3	-90	0	d_3	0

Step 11: Calculate the transformation matrices between frames based on:

$${}^{i-1}_iT = \begin{bmatrix} c\theta_i & -s\theta_i & 0 & a_{i-1} \\ c\alpha_{i-1}s\theta_i & c\alpha_{i-1}c\theta_i & -s\alpha_{i-1} & -s\alpha_{i-1}d_i \\ s\alpha_{i-1}s\theta_i & s\alpha_{i-1}c\theta_i & c\alpha_{i-1} & c\alpha_{i-1}d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$${}^0_1T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^1_2T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}, {}^2_3T = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{aligned}
 {}^0_3T &= {}^0_1T \cdot {}^1_2T \cdot {}^2_3T = \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\
 &= \begin{bmatrix} c\theta_1 & -s\theta_1 & 0 & 0 \\ s\theta_1 & c\theta_1 & 0 & 0 \\ 0 & 0 & 1 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} c_1 & 0 & -s_1 & -d_3 \cdot s_1 \\ s_1 & 0 & c_1 & d_3 \cdot c_1 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix}
 \end{aligned}$$

[2 marks for steps 1-5; 1.5 marks each for steps 6-9; 2 marks for step 10-11].

- (b) The end-effector (point P) is at a distance of L_3 from the origin of frame {3}. Derive the linear Jacobian matrix of the end-effector, with respect to the robot's frame {0}.

[5 marks]

The position of end-effector with respect to frame {3} is:

$${}^3P = \begin{bmatrix} 0 \\ 0 \\ L_3 \end{bmatrix}$$

Therefore, the position of end-effector with respect to frame {0} is:

$${}^0P = {}^0_3T \cdot {}^3P = \begin{bmatrix} c_1 & 0 & -s_1 & -d_3 \cdot s_1 \\ s_1 & 0 & c_1 & d_3 \cdot c_1 \\ 0 & -1 & 0 & d_2 \\ 0 & 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ L_3 \\ 1 \end{bmatrix} = \begin{bmatrix} -L_3 s_1 - d_3 \cdot s_1 \\ L_3 c_1 + d_3 \cdot c_1 \\ d_2 \\ 1 \end{bmatrix}$$

That means,

$$\begin{aligned}
 x &= -L_3 s_1 - d_3 \cdot s_1 \\
 y &= L_3 c_1 + d_3 \cdot c_1 \\
 z &= d_2
 \end{aligned}$$

The Jacobian matrix can be calculated from:

$$J = \begin{bmatrix} \frac{\partial x}{\partial \theta_1} & \frac{\partial x}{\partial d_2} & \frac{\partial x}{\partial d_3} \\ \frac{\partial y}{\partial \theta_1} & \frac{\partial y}{\partial d_2} & \frac{\partial y}{\partial d_3} \\ \frac{\partial z}{\partial \theta_1} & \frac{\partial z}{\partial d_2} & \frac{\partial z}{\partial d_3} \end{bmatrix} = \begin{bmatrix} -L_3 c_1 - d_3 c_1 & 0 & -s_1 \\ -L_3 s_1 - d_3 s_1 & 0 & c_1 \\ 0 & 1 & 0 \end{bmatrix}$$

[2 marks for calculating 0P , 3 marks for calculating Jacobian].

3. An RP robot is shown in Figure 3.1.

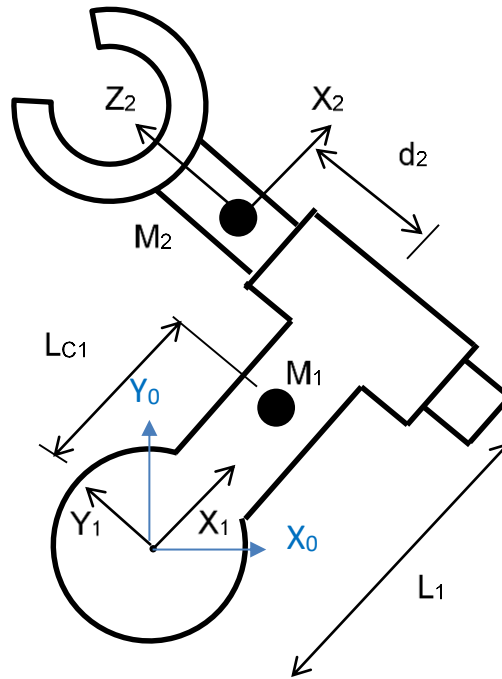


Figure 3.1: RP Robot

Frames $\{0\}$, $\{1\}$ and $\{2\}$, and all the geometrical dimensions of the robot are sketched in Figure 3.1 above. Gravitational pull is in the negative Y_0 direction. M_1 and M_2 are the masses of each link, and the big dots represent the centres of mass. Also, the inertia tensor of the two links with respect to their centres of mass are:

$${}^{c1}I_1 = \begin{bmatrix} I_{XX1} & 0 & 0 \\ 0 & I_{YY1} & 0 \\ 0 & 0 & I_{ZZ1} \end{bmatrix}$$

$${}^{c2}I_2 = \begin{bmatrix} I_{XX2} & 0 & 0 \\ 0 & I_{YY2} & 0 \\ 0 & 0 & I_{ZZ2} \end{bmatrix}$$

The rotation matrices for the robot are as follows:

$${}^0_1R = \begin{bmatrix} c_1 & -s_1 & 0 \\ s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$${}^1_2R = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$$

(a) Perform the outward iteration of the iterative Newton-Euler formula, and show that

$${}^1F_1 = \begin{bmatrix} m_1 s_1 g - m_1 L_{c1} \dot{\theta}_1^2 \\ m_1 c_1 g + m_1 L_{c1} \ddot{\theta}_1 \\ 0 \end{bmatrix}$$

$${}^1N_1 = \begin{bmatrix} 0 \\ 0 \\ I_{ZZ1} \ddot{\theta}_1 \end{bmatrix}$$

$${}^2F_2 = \begin{bmatrix} m_2 s_1 g - 2m_2 \dot{\theta}_1 \dot{d}_2 - m_2 d_2 \ddot{\theta}_1 - m_2 L_1 \dot{\theta}_1^2 \\ 0 \\ m_2 c_1 g + m_2 \ddot{d}_2 + m_2 L_1 \ddot{\theta}_1 - m_2 d_2 \dot{\theta}_1^2 \end{bmatrix}$$

$${}^2N_2 = \begin{bmatrix} 0 \\ -I_{YY2} \ddot{\theta}_1 \\ 0 \end{bmatrix}$$

[10 marks]

Preparation: Position vectors of frames and centre of mass:

$${}^1P_2 = [L_1 \quad d_2 \quad 0]^T$$

$${}^1P_{C1} = [L_{c1} \quad 0 \quad 0]^T$$

$${}^2P_{C2} = [0 \quad 0 \quad 0]^T$$

We also specify velocity and acceleration of the {0} frame.

$${}^0\omega_0 = 0$$

$${}^0\dot{\omega}_0 = 0$$

$${}^0\dot{v}_0 = [0 \quad g \quad 0]^T$$

Note that ${}^0\dot{v}_0$ has a g component in direction opposite to actual gravitational force.

Now we can start with the outward iteration for frame {1}:

$${}^1\omega_1 = \left({}^1_0R \cdot \underbrace{{}^0\omega_0}_0 \right) + (\dot{\theta}_1 \cdot {}^1Z_1) = \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix}$$

$${}^1\dot{\omega}_1 = \left({}^1_0R \cdot \underbrace{{}^0\dot{\omega}_0}_0 \right) + \left({}^1_0R \cdot \underbrace{{}^0\omega_0}_0 \times \dot{\theta}_1 \cdot {}^1Z_1 \right) + (\ddot{\theta}_1 \cdot {}^1Z_1) = \begin{bmatrix} 0 \\ 0 \\ \ddot{\theta}_1 \end{bmatrix}$$

$$\begin{aligned}
 {}^1\dot{v}_1 &= \left(\underbrace{{}^1R}_{{}^0R^T} \cdot {}^0\dot{v}_0 \right) + \left(2 \cdot {}^1\omega_1 \times \underbrace{\dot{d}_1}_0 \cdot {}^1Z_1 \right) + \left(\underbrace{\ddot{d}_1}_0 \cdot {}^1Z_1 \right) \\
 &\quad + {}^1_0R \left(\underbrace{{}^0\dot{\omega}_0}_0 \times {}^0P_1 + \underbrace{{}^0\omega_0}_0 \times {}^0\omega_0 \times {}^0P_1 \right) \\
 &= \begin{bmatrix} c_1 & s_1 & 0 \\ -s_1 & c_1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ g \\ 0 \end{bmatrix} = \begin{bmatrix} s_1 g \\ c_1 g \\ 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 {}^1\dot{v}_{C1} &= ({}^1\dot{v}_1) + ({}^1\dot{\omega}_1 \times {}^1P_{C1}) + ({}^1\omega_1 \times ({}^1\omega_1 \times {}^1P_{C1})) \\
 &= \begin{bmatrix} s_1 g \\ c_1 g \\ 0 \end{bmatrix} + \begin{bmatrix} i & j & k \\ 0 & 0 & \ddot{\theta}_1 \\ L_{C1} & 0 & 0 \end{bmatrix} \times \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{bmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ L_{C1} & 0 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} s_1 g \\ c_1 g \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ L_{C1} \ddot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ 0 & L_{C1} \dot{\theta}_1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} s_1 g - L_{C1} \dot{\theta}_1^2 \\ c_1 g + L_{C1} \ddot{\theta}_1 \\ 0 \end{bmatrix}
 \end{aligned}$$

$${}^1F_1 = m_1 \cdot {}^1\dot{v}_{C1} = \begin{bmatrix} m_1 s_1 g - m_1 L_{C1} \dot{\theta}_1^2 \\ m_1 c_1 g + m_1 L_{C1} \ddot{\theta}_1 \\ 0 \end{bmatrix}$$

$$\begin{aligned}
 {}^1N_1 &= {}^{c1}I_1 \cdot {}^1\dot{\omega}_1 + {}^1\omega_1 \times {}^{c1}I_1 \cdot {}^1\omega_1 \\
 &= \begin{bmatrix} I_{XX1} & 0 & 0 \\ 0 & I_{YY1} & 0 \\ 0 & 0 & I_{ZZ1} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{bmatrix} I_{XX1} & 0 & 0 \\ 0 & I_{YY1} & 0 \\ 0 & 0 & I_{ZZ1} \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ 0 \\ I_{ZZ1} \dot{\theta}_1 \end{bmatrix} + \begin{bmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ 0 & 0 & I_{ZZ1} \dot{\theta}_1 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ 0 \\ I_{ZZ1} \ddot{\theta}_1 \end{bmatrix}
 \end{aligned}$$

The outward iteration for frame {1} is complete. We now move on to frame {2}.

$${}^2\omega_2 = \left(\underbrace{{}^2R}_{{}^1R^T} \cdot {}^1\omega_1 \right) + \left(\underbrace{\dot{\varphi}_2}_0 \cdot {}^2Z_2 \right) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} = \begin{bmatrix} 0 \\ -\dot{\theta}_1 \\ 0 \end{bmatrix}$$

$$\begin{aligned}
 {}^2\dot{\omega}_2 &= ({}^2R \cdot {}^1\dot{\omega}_1) + \left({}^2R \cdot {}^1\omega_1 \times \underbrace{\dot{\varphi}_2}_0 \cdot {}^2Z_2 \right) + \left(\underbrace{\ddot{\varphi}_2}_0 \cdot {}^2Z_2 \right) \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} = \begin{bmatrix} 0 \\ -\ddot{\theta}_1 \\ 0 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 {}^2\dot{v}_2 &= ({}^2_1R \cdot {}^1\dot{v}_1) + (2 \cdot {}^2\omega_2 \times \dot{d}_2 \cdot {}^2Z_2) + (\ddot{d}_2 \cdot {}^2Z_2) \\
 &\quad + {}^2_1R({}^1\dot{\omega}_1 \times {}^1P_2 + {}^1\omega_1 \times {}^1\omega_1 \times {}^1P_2) \\
 &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \cdot \begin{bmatrix} s_1 g \\ c_1 g \\ 0 \end{bmatrix} + 2 \begin{bmatrix} i & j & k \\ 0 & -\dot{\theta}_1 & 0 \\ 0 & 0 & \dot{d}_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{d}_2 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \left\{ \begin{bmatrix} i & j & k \\ 0 & 0 & \ddot{\theta}_1 \\ L_1 & d_2 & 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \dot{\theta}_1 \end{bmatrix} \times \begin{bmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ L_1 & d_2 & 0 \end{bmatrix} \right\} \\
 &= \begin{bmatrix} s_1 g \\ 0 \\ c_1 g \end{bmatrix} + 2 \begin{bmatrix} -\dot{\theta}_1 \dot{d}_2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{d}_2 \end{bmatrix} \\
 &\quad + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \left\{ \begin{bmatrix} -d_2 \ddot{\theta}_1 \\ L_1 \ddot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} i & j & k \\ 0 & 0 & \dot{\theta}_1 \\ -d_2 \dot{\theta}_1 & L_1 \dot{\theta}_1 & 0 \end{bmatrix} \right\} \\
 &= \begin{bmatrix} s_1 g \\ 0 \\ c_1 g \end{bmatrix} + 2 \begin{bmatrix} -\dot{\theta}_1 \dot{d}_2 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ \ddot{d}_2 \end{bmatrix} + \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} -d_2 \ddot{\theta}_1 - L_1 \dot{\theta}_1^2 \\ L_1 \ddot{\theta}_1 - d_2 \dot{\theta}_1^2 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} s_1 g - 2\dot{\theta}_1 \dot{d}_2 - d_2 \ddot{\theta}_1 - L_1 \dot{\theta}_1^2 \\ 0 \\ c_1 g + \ddot{d}_2 + L_1 \ddot{\theta}_1 - d_2 \dot{\theta}_1^2 \end{bmatrix}
 \end{aligned}$$

$$\begin{aligned}
 {}^2\dot{v}_{C2} &= ({}^2\dot{v}_2) + \left({}^2\dot{\omega}_2 \times \underbrace{{}^2P_{C2}}_0 \right) + \left({}^2\omega_2 \times \left({}^2\omega_2 \times \underbrace{{}^2P_{C2}}_0 \right) \right) \\
 &= \begin{bmatrix} s_1 g - 2\dot{\theta}_1 \dot{d}_2 - d_2 \ddot{\theta}_1 - L_1 \dot{\theta}_1^2 \\ 0 \\ c_1 g + \ddot{d}_2 + L_1 \ddot{\theta}_1 - d_2 \dot{\theta}_1^2 \end{bmatrix}
 \end{aligned}$$

$${}^2F_2 = m_2 \cdot {}^2\dot{v}_{C2} = \begin{bmatrix} m_2 s_1 g - 2m_2 \dot{\theta}_1 \dot{d}_2 - m_2 d_2 \ddot{\theta}_1 - m_2 L_1 \dot{\theta}_1^2 \\ 0 \\ m_2 c_1 g + m_2 \ddot{d}_2 + m_2 L_1 \ddot{\theta}_1 - m_2 d_2 \dot{\theta}_1^2 \end{bmatrix}$$

$$\begin{aligned}
 {}^2N_2 &= {}^{C2}I_2 \cdot {}^2\dot{\omega}_2 + {}^2\omega_2 \times {}^{C2}I_2 \cdot {}^2\omega_2 \\
 &= \begin{bmatrix} I_{XX2} & 0 & 0 \\ 0 & I_{YY2} & 0 \\ 0 & 0 & I_{ZZ2} \end{bmatrix} \begin{bmatrix} 0 \\ -\ddot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} 0 \\ -\dot{\theta}_1 \\ 0 \end{bmatrix} \times \begin{bmatrix} I_{XX2} & 0 & 0 \\ 0 & I_{YY2} & 0 \\ 0 & 0 & I_{ZZ2} \end{bmatrix} \begin{bmatrix} 0 \\ -\dot{\theta}_1 \\ 0 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ -I_{YY2} \ddot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} i & j & k \\ 0 & -\dot{\theta}_1 & 0 \\ 0 & -I_{YY2} \dot{\theta}_1 & 0 \end{bmatrix} \\
 &= \begin{bmatrix} 0 \\ -I_{YY2} \ddot{\theta}_1 \\ 0 \end{bmatrix}
 \end{aligned}$$

[5 marks for link 1, 5 marks for link 2]

- (b) Perform the inward iteration of the iterative Newton-Euler formula to obtain the dynamic equations of the robot. Write the dynamic equations in the $M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \tau$ form. [10 marks]

Start from ${}^3f_3 = 0$, ${}^3n_3 = 0$

$${}^2f_2 = \left({}^2_3R \cdot \underbrace{{}^3f_3}_0 \right) + {}^2F_2 = \begin{bmatrix} m_2s_1g - 2m_2\dot{\theta}_1\dot{d}_2 - m_2d_2\ddot{\theta}_1 - m_2L_1\dot{\theta}_1^2 \\ 0 \\ m_2c_1g + m_2\ddot{d}_2 + m_2L_1\ddot{\theta}_1 - m_2d_2\dot{\theta}_1^2 \end{bmatrix}$$

$${}^2n_2 = \left({}^2_3R \cdot \underbrace{{}^3n_3}_0 \right) + \left(\underbrace{{}^2P_{C2}}_0 \times {}^2F_2 \right) + \left({}^2P_3 \times {}^2_3R \cdot \underbrace{{}^3f_3}_0 \right) + {}^2N_2 = \begin{bmatrix} 0 \\ -I_{YY2}\ddot{\theta}_1 \\ 0 \end{bmatrix}$$

${}^1f_1 =$ not important because the joint is revolute

$$\begin{aligned} {}^1n_1 &= ({}^1_2R \cdot {}^2n_2) + ({}^1P_{C1} \times {}^1F_1) + ({}^1P_2 \times {}^1_2R \cdot {}^2f_2) + {}^1N_1 \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ -I_{YY2}\ddot{\theta}_1 \\ 0 \end{bmatrix} + \begin{bmatrix} i & j & k \\ L_{C1} & 0 & 0 \\ {}^1F_{1x} & {}^1F_{1y} & {}^1F_{1z} \end{bmatrix} + \begin{bmatrix} L_1 \\ d_2 \\ 0 \end{bmatrix} \times \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix} \begin{bmatrix} {}^2f_{2x} \\ {}^2f_{2y} \\ {}^2f_{2z} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ I_{ZZ1}\ddot{\theta}_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ I_{YY2}\ddot{\theta}_1 \end{bmatrix} + \begin{bmatrix} 0 \\ -L_{C1} {}^1F_{1z} \\ L_{C1} {}^1F_{1y} \end{bmatrix} + \begin{bmatrix} i & j & k \\ L_1 & d_2 & 0 \\ {}^2f_{2x} & {}^2f_{2z} & -{}^2f_{2y} \end{bmatrix} + \begin{bmatrix} 0 \\ 0 \\ I_{ZZ1}\ddot{\theta}_1 \end{bmatrix} \\ &= \begin{bmatrix} 0 \\ 0 \\ I_{YY2}\ddot{\theta}_1 + L_{C1} {}^1F_{1y} + L_1 {}^2f_{2z} - d_2 {}^2f_{2x} + I_{ZZ1}\ddot{\theta}_1 \end{bmatrix} \end{aligned}$$

Dynamic equations:

$$\begin{aligned} \tau_1 &= \text{3rd row of } {}^1n_1 \\ &= I_{YY2}\ddot{\theta}_1 + L_{C1}(m_1c_1g + m_1L_{C1}\ddot{\theta}_1) + L_1(m_2c_1g + m_2\ddot{d}_2 + m_2L_1\ddot{\theta}_1 - m_2d_2\dot{\theta}_1^2) \\ &\quad - d_2(m_2s_1g - 2m_2\dot{\theta}_1\dot{d}_2 - m_2d_2\ddot{\theta}_1 - m_2L_1\dot{\theta}_1^2) + I_{ZZ1}\ddot{\theta}_1 \end{aligned}$$

$$\begin{aligned} \tau_2 &= \text{3rd row of } {}^2f_2 \\ &= m_2c_1g + m_2\ddot{d}_2 + m_2L_1\ddot{\theta}_1 - m_2d_2\dot{\theta}_1^2 \end{aligned}$$

In $M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \tau$ form:

$$\begin{bmatrix} I_{YY2} + m_1L_{C1}^2 + m_2L_1^2 + m_2d_2^2 + I_{ZZ1} & m_2L_1 \\ m_2L_1 & m_2 \end{bmatrix} \begin{bmatrix} \ddot{\theta}_1 \\ \ddot{d}_2 \end{bmatrix} + \begin{bmatrix} -m_2L_1d_2\dot{\theta}_1^2 + 2m_2d_2\dot{\theta}_1\dot{d}_2 + m_2L_1d_2\dot{\theta}_1^2 \\ -m_2d_2\dot{\theta}_1^2 \end{bmatrix} + \begin{bmatrix} m_1L_{C1}c_1g + m_2L_1c_1g - m_2d_2s_1g \\ m_2c_1g \end{bmatrix} = \begin{bmatrix} \tau_1 \\ \tau_2 \end{bmatrix}$$

[5 marks for terms before dynamic equations, 5 marks for dynamic equations]

4. A two-link planar RR robot is required to move from $(x = 4 \text{ m}, y = 3 \text{ m})$ to $(x = 7 \text{ m}, y = 1 \text{ m})$ within 5 seconds, following a “linear functions with parabolic blend” trajectory.

- (a) What are the minimum accelerations of the blend region in x -direction and y -direction, so that the linear region will exist? If you can only set the same acceleration for both directions, which value would you choose and why?

[3 marks]

The acceleration must satisfy $\ddot{u} \geq 4 \frac{|u_f - u_0|}{t_f^2}$.

For x -direction, this is $\ddot{u} \geq 4 \frac{|7-4|}{5^2} \text{ m} \cdot \text{s}^{-2} = 0.48 \text{ m} \cdot \text{s}^{-2}$.

For y -direction, this is $\ddot{u} \geq 4 \frac{|1-3|}{5^2} \text{ m} \cdot \text{s}^{-2} = 0.32 \text{ m} \cdot \text{s}^{-2}$.

I will choose the larger of the two, i.e. $0.48 \text{ m} \cdot \text{s}^{-2}$. Otherwise, there will be no linear region for x -direction.

[1 mark for x , 1 mark for y , 1 mark for the choice of single acceleration]

- (b) Assume that you have chosen the acceleration of the blend region to be $1 \text{ m} \cdot \text{s}^{-2}$ for both x -direction and y -direction. Derive the equations of the trajectories in x -direction and y -direction.

[12 marks]

For x -direction:

The time at the end of blend region is:

$$\begin{aligned} t_b &= \frac{t_f}{2} - \frac{\sqrt{\ddot{u}^2 t_f^2 - 4\ddot{u}(u_f - u_0)}}{2\ddot{u}} \\ &= \frac{5}{2} - \frac{\sqrt{1 \cdot 5^2 - 4 \cdot 1 \cdot (7 - 4)}}{2 \cdot 1} \\ &= \frac{5}{2} - \frac{\sqrt{25 - 12}}{2} \\ &= 0.6972 \text{ s} \end{aligned}$$

The x -position at the end of blend region is:

$$u_b = u_0 + \frac{1}{2} \ddot{u} t_b^2 = 4 + \frac{1}{2} \cdot 1 \cdot 0.6972^2 = 4.2431 \text{ m}$$

The midpoint of the x -trajectory is:

$$u_h = \frac{u_0 + u_f}{2} = \frac{4 + 7}{2} = 5.5 \text{ m}$$

The gradient of the linear portion is:

$$\frac{u_h - u_b}{t_h - t_b} = \frac{5.5 - 4.2431}{2.5 - 0.6972} \text{ m}\cdot\text{s}^{-1} = 0.6972 \text{ m}\cdot\text{s}^{-1}$$

The x -trajectory is then:

$$u(t) = \begin{cases} 4 \text{ m} + \frac{1}{2} \cdot 1 \cdot \text{m}\cdot\text{s}^{-2} \cdot t^2 & t < 0.6972 \text{ s} \\ 0.6972 \cdot \text{m}\cdot\text{s}^{-1} \cdot (t - 0.6972 \text{ s}) + 4.2431 \text{ m} & 0.6972 \text{ s} \leq t < 4.3028 \text{ s} \\ 7 \text{ m} - \frac{1}{2} \cdot 1 \cdot \text{m}\cdot\text{s}^{-2} \cdot (5 \text{ s} - t)^2 & t \geq 4.3028 \text{ s} \end{cases}$$

For y -direction:

The time at the end of blend region is:

$$\begin{aligned} t_b &= \frac{t_f}{2} - \frac{\sqrt{\ddot{u}^2 t_f^2 - 4\ddot{u}(u_f - u_0)}}{2\ddot{u}} \\ &= \frac{5}{2} - \frac{\sqrt{1 \cdot 5^2 - 4 \cdot 1 \cdot |1 - 3|}}{2 \cdot 1} \\ &= \frac{5}{2} - \frac{\sqrt{25 - 8}}{2} \\ &= 0.4384 \text{ s} \end{aligned}$$

The y -position at the end of blend region is (minus because in negative direction):

$$u_b = u_0 - \frac{1}{2} \ddot{u} t_b^2 = 3 - \frac{1}{2} \cdot 1 \cdot 0.4384^2 = 2.9039 \text{ m}$$

The midpoint of the y -trajectory is:

$$u_h = \frac{u_0 + u_f}{2} = \frac{1 + 3}{2} = 2 \text{ m}$$

The gradient of the linear portion is:

$$\frac{u_h - u_b}{t_h - t_b} = \frac{2 - 2.9039}{2.5 - 0.4384} \text{ m}\cdot\text{s}^{-1} = -0.4384 \text{ m}\cdot\text{s}^{-1}$$

The y -trajectory is then:

$$u(t) = \begin{cases} 3 \text{ m} - \frac{1}{2} \cdot 1 \cdot \text{m}\cdot\text{s}^{-2} \cdot t^2 & t < 0.4384 \text{ s} \\ -0.4387 \cdot \text{m}\cdot\text{s}^{-1} \cdot (t - 0.4387 \text{ s}) + 2.9039 \text{ m} & 0.4384 \text{ s} \leq t < 4.5616 \text{ s} \\ 1 \text{ m} + \frac{1}{2} \cdot 1 \cdot \text{m}\cdot\text{s}^{-2} \cdot (5 \text{ s} - t)^2 & t \geq 4.5616 \text{ s} \end{cases}$$

[For x : 4 points for u_b, t_b and gradient. 2 points for putting in final form. Same for y]

- (c) The joint-space dynamic model of the robot is $M(q)\ddot{q} + V(q, \dot{q}) + G(q) = \tau - \tau_{\text{friction}}$. Derive a controller which will ensure that the robot follows the desired trajectory, with a stiffness of k_{p1} for joint 1 and k_{p2} for joint 2. The robot should be critically damped. Additionally, sketch the overall control structure (block diagram) of the robot, given the xy trajectories.

[5 marks]

Firstly, design a model-based compensator as:

$$\tau = M(q)f + V(q, \dot{q}) + G(q) + \tau_{\text{friction}}$$

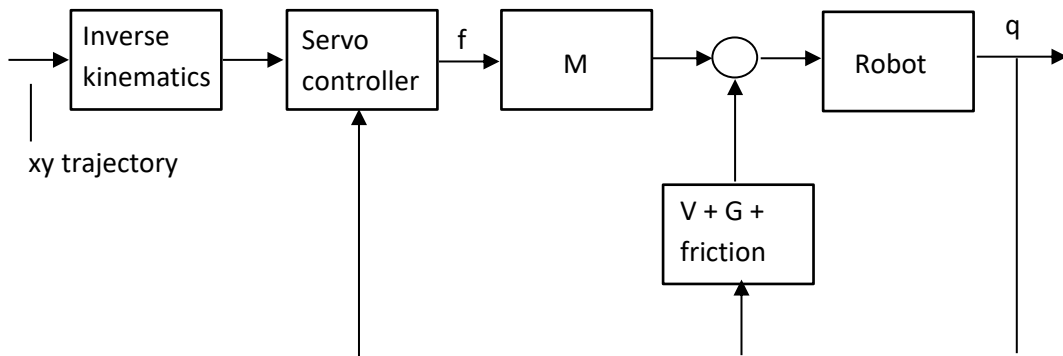
Then, design the servo controller as:

$$f = \ddot{q}_d - K_v(\dot{q} - \dot{q}_d) - K_p(q - q_d)$$

where:

$$K_p = \begin{bmatrix} k_{p1} & 0 \\ 0 & k_{p2} \end{bmatrix}, K_v = \begin{bmatrix} 2\sqrt{k_{p1}} & 0 \\ 0 & 2\sqrt{k_{p2}} \end{bmatrix}$$

Block diagram: Students should show an inverse kinematic block to transform xy trajectories to joint space trajectories.



[3 marks for equations, 2 marks for block diagram]

ELEC0140-A6U-2021-Solution

5. The black-and-white image of a rectangular object has the pixel values as shown in Table 5.1.
(Note: 1st row represents the x-coordinate of a pixel, 1st column represents the y-coordinate of a pixel. They are not part of the image!):

Table 5.1: Pixel values of a rectangular object.

y↓ x→	1	2	3	4	5	6	7	8	9	10
1	0	0	0	0	0	0	0	0	0	0
2	0	0	0	0	0	0	0	0	0	0
3	0	0	0	0	0	0	0	1	1	0
4	0	0	0	0	0	1	1	1	1	0
5	0	0	0	1	1	1	1	1	1	0
6	0	1	1	1	1	1	1	0	0	0
7	0	1	1	1	1	0	0	0	0	0
8	0	1	1	0	0	0	0	0	0	0
9	0	0	0	0	0	0	0	0	0	0
10	0	0	0	0	0	0	0	0	0	0

- (a) Use the “moment” method to find the centroid (leave your answer in 1 decimal point), and show that the inertia matrix of the object is $J = \begin{bmatrix} 126 & -60 \\ -60 & 46 \end{bmatrix}$.

[8 marks]

$$M_{00} = \sum I(x, y) = \text{sum of 1's in image} = 24$$

$$M_{10} = \sum x \cdot I(x, y) = (2 + 2 + 2) + (3 + 3 + 3) + (4 + 4 + 4) + \dots + (9 + 9 + 9) = 132$$

$$\begin{aligned} M_{01} &= \sum y \cdot I(x, y) = (3 + 3) + (4 + 4 + 4 + 4) + (5 + 5 + 5 + 5 + 5 + 5) \\ &\quad + (6 + 6 + 6 + 6 + 6 + 6) + (7 + 7 + 7 + 7) + (8 + 8) \\ &= 6 + 16 + 30 + 36 + 28 + 16 \\ &= 132 \end{aligned}$$

The centroid is:

$$X_c = \frac{M_{10}}{M_{00}} = \frac{132}{24} = 5.5; \quad Y_c = \frac{M_{01}}{M_{00}} = \frac{132}{24} = 5.5$$

$$\begin{aligned} M_{11} &= \sum xy \cdot I(xy) = 2 \cdot 6 + 2 \cdot 7 + 2 \cdot 8 + 3 \cdot 6 + 3 \cdot 7 + 3 \cdot 8 + 4 \cdot 5 + 4 \cdot 6 + 4 \cdot 7 \\ &\quad + 5 \cdot 5 + 5 \cdot 6 + 5 \cdot 7 + 6 \cdot 4 + 6 \cdot 5 + 6 \cdot 6 \\ &\quad + 7 \cdot 4 + 7 \cdot 5 + 7 \cdot 6 + 8 \cdot 3 + 8 \cdot 4 + 8 \cdot 5 \\ &\quad + 9 \cdot 3 + 9 \cdot 4 + 9 \cdot 5 \\ &= 12 + 14 + 16 + 18 + 21 + 24 + 20 + 24 + 28 \\ &\quad + 25 + 30 + 35 + 24 + 30 + 36 \\ &\quad + 28 + 35 + 42 + 24 + 32 + 40 \\ &\quad + 27 + 36 + 45 \\ &= 666 \end{aligned}$$

$$M_{20} = \sum x^2 \cdot I(x, y) = (4 + 4 + 4) + (9 + 9 + 9) + \dots + (81 + 81 + 81) = 852$$

$$\begin{aligned} M_{02} &= \sum y^2 \cdot I(x, y) = (9 + 9) + (16 + 16 + 16 + 16) + (25 + 25 + 25 + 25 + 25 + 25) \\ &\quad + (36 + 36 + 36 + 36 + 36 + 36) + (49 + 49 + 49 + 49) + (64 + 64) \\ &= 18 + 64 + 150 + 216 + 196 + 128 \\ &= 772 \end{aligned}$$

$$\mu_{11} = M_{11} - X_C M_{01} = 666 - 5.5 \cdot 132 = -60$$

$$\mu_{20} = M_{20} - X_C M_{10} = 852 - 5.5 \cdot 132 = 126$$

$$\mu_{02} = M_{02} - Y_C M_{01} = 772 - 5.5 \cdot 132 = 46$$

The inertia matrix is:

$$J = \begin{bmatrix} \mu_{20} & \mu_{11} \\ \mu_{11} & \mu_{02} \end{bmatrix} = \begin{bmatrix} 126 & -60 \\ -60 & 46 \end{bmatrix}$$

[3 marks for centroid, 5 marks for inertia matrix]

- (b) From the inertia matrix, calculate the major radius, minor radius and angle of the object. Let all the intermediate calculations and final answer to be in 1 decimal point. Note: Do not use MATLAB to obtain the answer.

[12 marks]

First, we need to calculate the eigenvalues of J . This can be done by equating the determinant of $(\lambda I - J)$ as zero:

$$|\lambda I - J| = 0$$

$$\begin{vmatrix} \lambda - 126 & 60 \\ 60 & \lambda - 46 \end{vmatrix} = (\lambda - 126)(\lambda - 46) - 3600 = 0$$

$$\lambda^2 - 172\lambda + 5796 - 3600 = 0$$

$$\lambda^2 - 172\lambda + 2196 = 0$$

$$\lambda = \frac{172 \pm \sqrt{172^2 - 4 \cdot 2196}}{2} = \frac{172 \pm \sqrt{20800}}{2} = 158.1 \text{ (158.1110) and } 13.9 \text{ (13.8890)}$$

We then calculate the eigenvector of J corresponding to the larger eigenvalue.

$$(J - \lambda I)v = 0$$

$$\begin{bmatrix} 126 - 158.1 & -60 \\ -60 & 46 - 158.1 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} -32.1 & -60 \\ -60 & -112.1 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$\begin{bmatrix} 32.1 & 60 \\ 60 & 112.1 \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

We shall ignore the trivial solutions here, and find the relationship between v_x and v_y .

$$\begin{aligned} 32.1v_x &= -60v_y \\ 60v_x &= -112.1v_y \end{aligned}$$

Assume $v_y = 1$, we thus have:

$$\begin{aligned} 32.1v_x &= -60 \\ 60v_x &= -112.1 \\ v_x &= \frac{-60}{32.1} = -1.9 \\ v_x &= \frac{-112.1}{60} = -1.9 \end{aligned}$$

The eigenvector is therefore

$$v = \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} -1.9 \\ 1 \end{bmatrix}$$

We can also scale this by its vector to have a unit magnitude: (Note this is not crucial, as the ratio of v_x and v_y is more important, not the individual values).

$$v = \begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} -0.9 \\ 0.5 \end{bmatrix} \left(= \begin{bmatrix} -0.8817 \\ 0.4719 \end{bmatrix} \right)$$

Now we can calculate the major radius, minor radius and angle of the object:

$$a = 2 \sqrt{\frac{\lambda_1}{M_{00}}} = 2 \sqrt{\frac{158.1}{24}} = 5.1$$

$$b = 2 \sqrt{\frac{\lambda_2}{M_{00}}} = 2 \sqrt{\frac{13.9}{24}} = 1.5$$

$$\theta = \tan^{-1} \left(\frac{v_y}{v_x} \right) = \tan^{-1} \left(\frac{0.5}{-0.9} \right) = -29.1^\circ$$

[4 marks for eigenvalues, 5 marks for eigenvectors, 3 marks for radius and angle]

6. Frame $\{B\}$ is initially coincident with frame $\{A\}$. First, $\{B\}$ is rotated about y_A by β degrees, and is subsequently rotated about x_A by α degrees, followed by a rotation about z_A by γ degrees.

(a) Show that the rotation matrix which accomplishes these 3 rotations in the given order is:

$${}^A_B R = \begin{bmatrix} c\gamma c\beta - s\gamma s\alpha s\beta & -s\gamma c\alpha & c\gamma s\beta + s\gamma s\alpha c\beta \\ s\gamma c\beta + c\gamma s\alpha s\beta & c\gamma c\alpha & s\gamma s\beta - c\gamma s\alpha c\beta \\ -c\alpha s\beta & s\alpha & c\alpha c\beta \end{bmatrix}$$

[3 marks]

The rotation is represented using the fixed angles representation, as the rotations occur about the original (fixed) axis. In fixed angle representation, the order of matrix multiplication is the opposite of the order of physical rotations:

Physical rotation: Y then X then Z.

Rotation matrix:

$$\begin{aligned} {}^A_B R &= R_Z(\gamma) \cdot R_X(\alpha) \cdot R_Y(\beta) \\ &= \begin{bmatrix} c\gamma & -s\gamma & 0 \\ s\gamma & c\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\alpha & -s\alpha \\ 0 & s\alpha & c\alpha \end{bmatrix} \cdot \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \\ &= \begin{bmatrix} c\gamma & -s\gamma c\alpha & s\gamma s\alpha \\ s\gamma & c\gamma c\alpha & -c\gamma s\alpha \\ 0 & s\alpha & c\alpha \end{bmatrix} \cdot \begin{bmatrix} c\beta & 0 & s\beta \\ 0 & 1 & 0 \\ -s\beta & 0 & c\beta \end{bmatrix} \\ &= \begin{bmatrix} c\gamma c\beta - s\gamma s\alpha s\beta & -s\gamma c\alpha & c\gamma s\beta + s\gamma s\alpha c\beta \\ s\gamma c\beta + c\gamma s\alpha s\beta & c\gamma c\alpha & s\gamma s\beta - c\gamma s\alpha c\beta \\ -c\alpha s\beta & s\alpha & c\alpha c\beta \end{bmatrix} \end{aligned}$$

[3 marks for steps and final answer]

- (b) If the above rotation matrix has the following numerical values,

$${}^A_B R = \begin{bmatrix} 0.6830 & -0.1830 & 0.7071 \\ 0.5657 & 0.7450 & -0.3536 \\ -0.4620 & 0.6414 & 0.6123 \end{bmatrix},$$

calculate the angles α , β and γ .

[7 marks]

We see that:

$$r_{31}^2 + r_{33}^2 = c^2\alpha \cdot s^2\beta + c^2\alpha \cdot c^2\beta = c^2\alpha$$

Hence:

$$c\alpha = \sqrt{r_{31}^2 + r_{33}^2}$$

To make sure we have the correct quadrant, use tangent instead of only cosine:

$$\tan \alpha = \frac{s\alpha}{c\alpha} = \frac{r_{32}}{\sqrt{r_{31}^2 + r_{33}^2}}$$

$$\alpha = \arctan2\left(r_{32}, \sqrt{r_{31}^2 + r_{33}^2}\right)$$

Then,

$$\tan \beta = \frac{s\beta}{c\beta} = \frac{-r_{31}/c\alpha}{r_{33}/c\alpha}$$

$$\beta = \arctan2(-r_{31}/c\alpha, r_{33}/c\alpha)$$

And,

$$\tan \gamma = \frac{s\gamma}{c\gamma} = \frac{-r_{12}/c\alpha}{r_{22}/c\alpha}$$

$$\gamma = \arctan2(-r_{12}/c\alpha, r_{22}/c\alpha)$$

Putting in the numerical values:

$$\alpha = \arctan2\left(r_{32}, \sqrt{r_{31}^2 + r_{33}^2}\right) = \arctan2\left(0.6414, \sqrt{0.4620^2 + 0.6123^2}\right) = 39.9^\circ$$

$$\beta = \arctan2(-r_{31}/c\alpha, r_{33}/c\alpha) = \arctan2\left(\frac{0.4620}{0.7671}, \frac{0.6123}{0.7671}\right) = 37.0^\circ$$

$$\gamma = \arctan2(-r_{12}/c\alpha, r_{22}/c\alpha) = \arctan2\left(\frac{0.1830}{0.7671}, \frac{0.7450}{0.7671}\right) = 13.8^\circ$$

[4 marks for derivation, 3 marks for numerical values]